

Thesis

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```
rm(list=ls())
data <- read.csv('data_with_pre_data.csv')
set.seed(67)
```

```
head(data)
```

##	X	PLAYER_ID	SEASON	PLAYER	PLAYER_SLUG	NUM	POSITION	WEIGHT	BIRTH_DATE	
##	1	0	1631210	2024	Jacob Toppin	jacob-toppin	0	F	200	5/8/2000
##	2	1	1630552	2024	Jalen Johnson	jalen-johnson	1	F	219	12/18/2001
##	3	2	1627747	2024	Caris LeVert	caris-levert	3	G	205	8/25/1994
##	4	3	1641723	2024	Kobe Bufkin	kobe-bufkin	4	G	195	9/21/2003
##	5	4	1629027	2024	Trae Young	trae-young	11	G	164	9/19/1998
##	6	5	1629611	2024	Terance Mann	terance-mann	14	G-F	215	10/18/1996
##	AGE	EXP	SCHOOL	HEIGHT_INCHES	NBA_GP	NBA_GS	NBA_MIN	NBA_FGM	NBA_FGA	
##	1	25	1	Kentucky	80	26	0	113	14	31
##	2	23	3	Duke	80	184	94	4335	810	1607
##	3	30	8	Michigan	78	524	223	14550	2692	6159
##	4	21	1	Michigan	76	27	0	319	52	139
##	5	26	6	Oklahoma	73	483	483	16621	3865	8935
##	6	28	5	Florida State	77	412	169	9101	1280	2549
##	NBA_FG_PCT	NBA_FG3M	NBA_FG3A	NBA_FG3_PCT	NBA_FTM	NBA_FTA	NBA_FT_PCT	NBA_OREB		
##	1	0.452	6	14	0.429	2	2	1.000	6	
##	2	0.504	148	458	0.323	255	358	0.712	189	
##	3	0.437	779	2257	0.345	1111	1509	0.736	369	
##	4	0.374	13	59	0.220	17	26	0.654	11	
##	5	0.433	1277	3628	0.352	3213	3679	0.873	299	
##	6	0.502	329	884	0.372	442	566	0.781	402	
##	NBA_DREB	NBA_REB	NBA_AST	NBA_STL	NBA_BLK	NBA_TOV	NBA_PF	NBA_PTS		
##	1	16	22	9	2	1	4	6	36	
##	2	966	1155	470	164	119	257	322	2023	
##	3	1612	1981	2105	548	198	918	1034	7274	
##	4	43	54	44	10	7	18	29	134	
##	5	1388	1687	4748	505	74	2051	841	12220	
##	6	1049	1451	790	227	91	317	772	3331	
##	NBA_FG_pct_inc	NBA_FG3_pct_inc	NBA_FT_pct_inc	PRE_G	PRE_GS	PRE_MP	PRE_FG			
##	1	0.000	0.000	0.000	116	40	2516	327		
##	2	-0.033	0.038	0.084	13	8	278	57		
##	3	-0.010	0.023	0.023	103	70	2721	366		
##	4	0.004	0.000	0.000	61	33	1419	203		
##	5	0.015	0.028	0.044	32	32	1133	261		
##	6	0.033	0.022	0.104	140	101	3620	505		

	PRE_FGA	PRE_FG_PCT	PRE_3P	PRE_3PA	PRE_3P_PCT	PRE_2P	PRE_2PA	PRE_2P_PCT		
## 1	695	0.471	39	135	0.289	288	560	0.514		
## 2	109	0.523	8	18	0.444	49	91	0.538		
## 3	843	0.434	134	334	0.401	232	509	0.456		
## 4	438	0.463	51	157	0.325	152	281	0.541		
## 5	618	0.422	118	328	0.360	143	290	0.493		
## 6	915	0.552	54	165	0.327	451	750	0.601		
	PRE_eFG_PCT	PRE_FT	PRE_FTA	PRE_FT_PCT	PRE_ORB	PRE_DRB	PRE_TRB	PRE_AST	PRE_STL	
## 1	0.499	171	246	0.695	156	363	519	137	48	
## 2	0.560	24	38	0.632	20	59	79	29	15	
## 3	0.514	204	265	0.770	39	325	364	275	97	
## 4	0.522	90	108	0.833	34	147	181	104	54	
## 5	0.518	236	274	0.861	14	111	125	279	54	
## 6	0.581	258	385	0.670	266	444	710	269	112	
	PRE_BLK	PRE_TOV	PRE_PF	PRE_PTS	PRE_FG_pct_inc	PRE_3P_pct_inc	PRE_2P_pct_inc			
## 1	50	103	199	864	0.130		0.060		0.104	
## 2	16	33	29	146	0.000		0.000		0.000	
## 3	22	136	151	1070	0.191		0.144		0.220	
## 4	26	79	104	547	0.102		0.133		0.034	
## 5	8	167	57	876	0.000		0.000		0.000	
## 6	34	191	284	1322	0.079		0.140		0.088	
	PRE_FT_pct_inc	PRE_COL_or_INT								
## 1	0.136		COL							
## 2	0.000		COL							
## 3	0.043		COL							
## 4	0.000		COL							
## 5	0.000		COL							
## 6	0.332		COL							

Normalize to Per Minute Data

```
normalize_stats <- function(df) {
  nba_start <- which(names(df) == "NBA_MIN") #split the data
  nba_end   <- which(names(df) == "PRE_G")

  for (i in (nba_start+1):(nba_end-1)) {
    colname <- names(df)[i]
    if (!grepl("PCT", colname, ignore.case = TRUE) && is.numeric(df[[colname]])) { #divide total column
      df[[colname]] <- round(as.numeric(df[[colname]]) / as.numeric(df[["NBA_MIN"]]),3)
    }
  }
  pre_start <- which(names(df) == "PRE_MP")
  for (i in (pre_start+1):ncol(df)) { #repeat for college
    colname <- names(df)[i]
    if (!grepl("PCT", colname, ignore.case = TRUE) && is.numeric(df[[colname]])) {
      df[[colname]] <- round(as.numeric(df[[colname]]) / as.numeric(df[["PRE_MP"]]),3)
    }
  }
  return(df)
}
per_min_data <- normalize_stats(data)
```

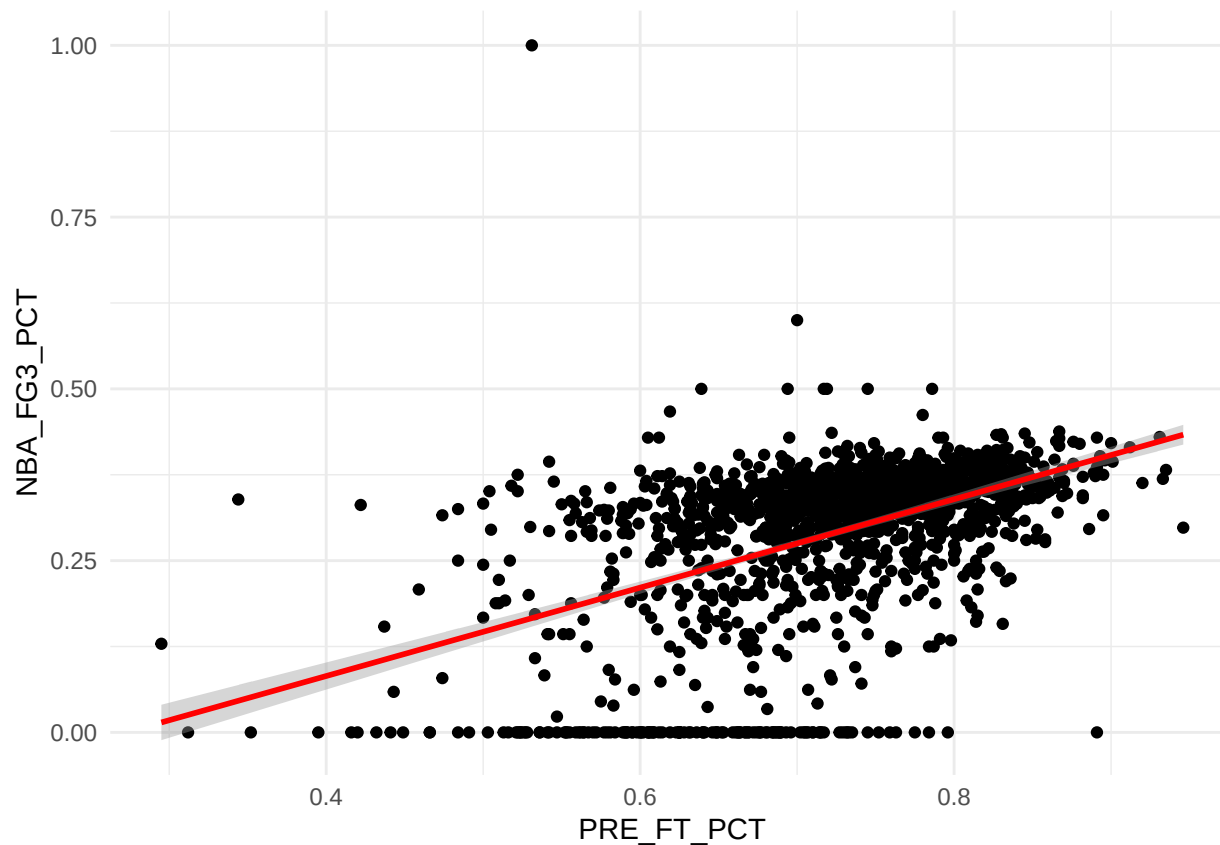
Simple Linear Model Predicting 3 Point Percentage from Pre-NBA Free Throw Percentage

```
FTto3 <- lm(NBA_FG3_PCT ~ PRE_FT_PCT, data = data)
summary(FTto3)
```

```
##
## Call:
## lm(formula = NBA_FG3_PCT ~ PRE_FT_PCT, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.39789 -0.03519  0.02157  0.06188  0.83370
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.17531    0.02196  -7.984 2.83e-15 ***
## PRE_FT_PCT   0.64333    0.03026  21.262 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1028 on 1477 degrees of freedom
## Multiple R-squared:  0.2343, Adjusted R-squared:  0.2338
## F-statistic: 452.1 on 1 and 1477 DF,  p-value: < 2.2e-16
```

```
ggplot(data, aes(x = PRE_FT_PCT, y = NBA_FG3_PCT)) +
  geom_point() +
  geom_smooth(method = "lm", color = "red") +
  labs(x = "PRE_FT_PCT", y = "NBA_FG3_PCT") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
data_filtered <- data %>%
  filter(PRE_FTA > 25 & NBA_FG3A > 25)

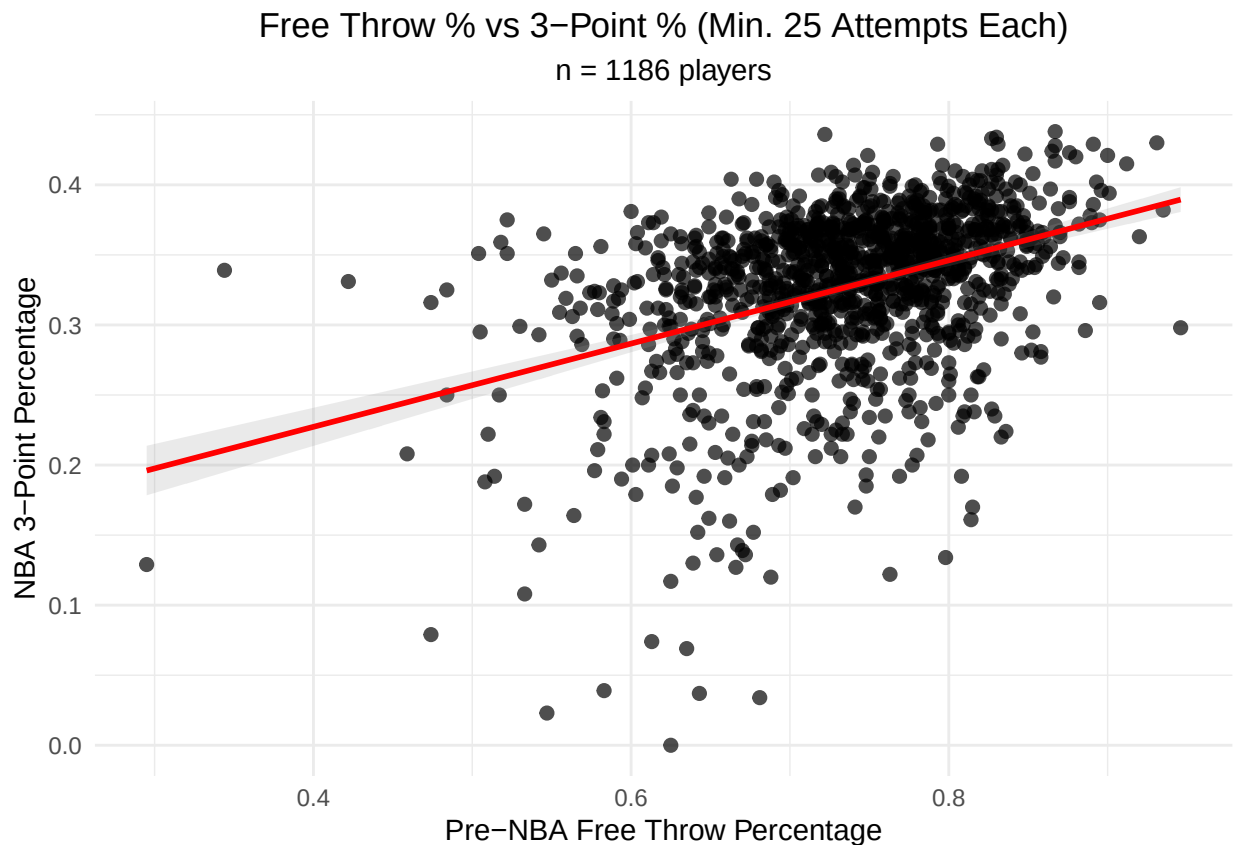
data_filtered <- normalize_stats(data_filtered)

FTto3 <- lm(NBA_FG3_PCT ~ PRE_FT_PCT, data = data_filtered)
summary(FTto3)
```

```
##
## Call:
## lm(formula = NBA_FG3_PCT ~ PRE_FT_PCT, data = data_filtered)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.29412 -0.02186  0.01026  0.03692  0.12836
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.10845    0.01488   7.287 5.77e-13 ***
## PRE_FT_PCT   0.29707    0.02010  14.783 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.05433 on 1184 degrees of freedom
## Multiple R-squared:  0.1558, Adjusted R-squared:  0.1551
## F-statistic: 218.5 on 1 and 1184 DF, p-value: < 2.2e-16
```

```
ggplot(data_filtered, aes(x = PRE_FT_PCT, y = NBA_FG3_PCT)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "red", alpha = 0.2) +
  labs(
    x = "Pre-NBA Free Throw Percentage",
    y = "NBA 3-Point Percentage",
    title = "Free Throw % vs 3-Point % (Min. 25 Attempts Each)",
    subtitle = paste("n =", nrow(data_filtered), "players")
  ) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(hjust = 0.5))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Simple Linear Model Predicting 3 Point Percentage from Pre-NBA 3 Point Percentage

```
Pre3toNBA3 <- lm(NBA_FG3_PCT ~ PRE_3P_PCT, data = data)
summary(Pre3toNBA3)
```

```
##
## Call:
## lm(formula = NBA_FG3_PCT ~ PRE_3P_PCT, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.53015 -0.02247  0.02799  0.05813  0.70037
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.184544   0.007975   23.14  <2e-16 ***
## PRE_3P_PCT   0.345604   0.023060   14.99  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1012 on 1407 degrees of freedom
## (70 observations deleted due to missingness)
## Multiple R-squared:  0.1377, Adjusted R-squared:  0.1371
## F-statistic: 224.6 on 1 and 1407 DF,  p-value: < 2.2e-16
```

```
ggplot(data_filtered, aes(x = PRE_3P_PCT, y = NBA_FG3_PCT, color = PRE_COL_or_INT)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "black", alpha = 0.05) +
  labs(
    x = "Pre-NBA Three Point Percentage",
    y = "NBA 3-Point Percentage",
    title = "Pre-NBA 3-Point % vs NBA 3-Point % (Min. 25 Attempts Each)",
    subtitle = paste("n =", nrow(data_filtered), "players")
  ) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        plot.subtitle = element_text(hjust = 0.5))
```

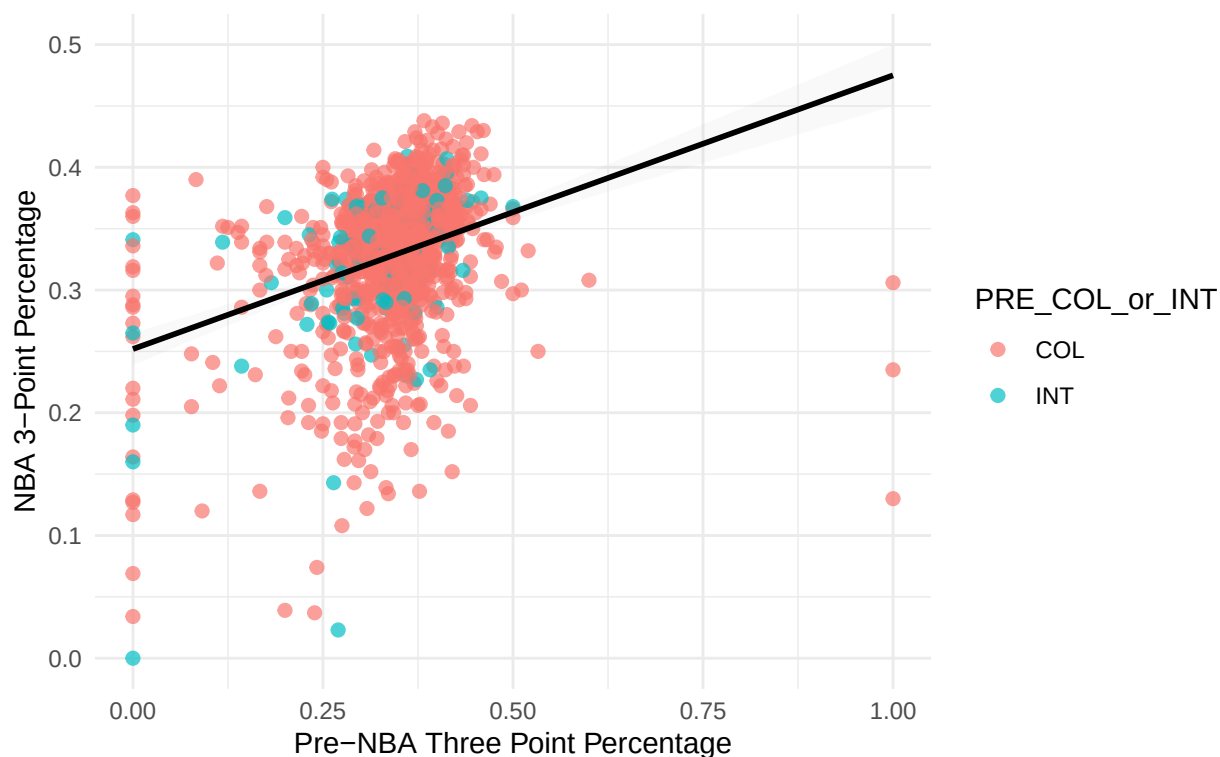
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 13 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 13 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Pre-NBA 3-Point % vs NBA 3-Point % (Min. 25 Attempts Each)

n = 1186 players



#ANOVA INTERNATIONAL VS COLLEGE # MLR of both PRE-NBA FT% and 3P% vs NBA 3P%

```
FT3P_to_NBA3 <- lm(NBA_FG3_PCT ~ PRE_3P_PCT + PRE_FT_PCT, data=data)
summary(FT3P_to_NBA3)
```

```
##
## Call:
## lm(formula = NBA_FG3_PCT ~ PRE_3P_PCT + PRE_FT_PCT, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.42297 -0.02671  0.01852  0.05416  0.79144
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.11095    0.02246  -4.940 8.74e-07 ***
## PRE_3P_PCT   0.22633    0.02325   9.735 < 2e-16 ***
## PRE_FT_PCT   0.45976    0.03295  13.953 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09489 on 1406 degrees of freedom
## (70 observations deleted due to missingness)
## Multiple R-squared:  0.2426, Adjusted R-squared:  0.2415
## F-statistic: 225.1 on 2 and 1406 DF, p-value: < 2.2e-16
```

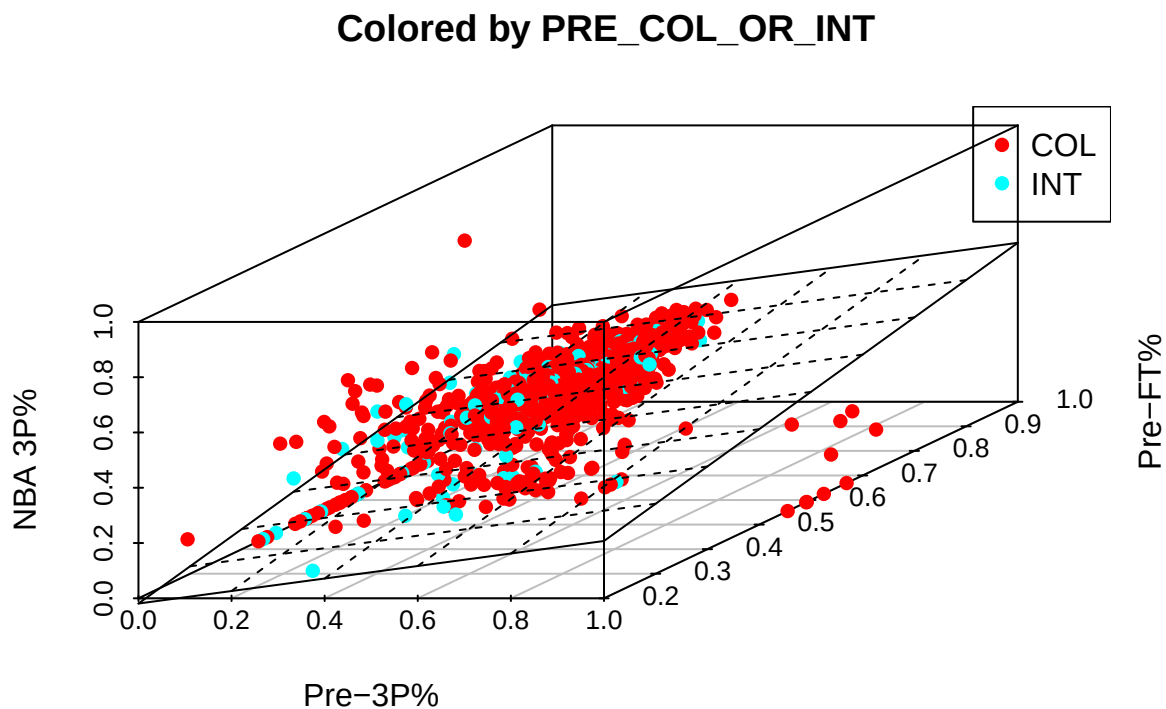
```

clean_data <- na.omit(data[c("NBA_FG3_PCT", "PRE_3P_PCT", "PRE_FT_PCT", "PRE_COL_or_INT")])
model <- lm(NBA_FG3_PCT ~ PRE_3P_PCT + PRE_FT_PCT, data = clean_data)

color_factor <- as.factor(clean_data$PRE_COL_or_INT)
colors <- rainbow(length(levels(color_factor)))[color_factor]
s3d <- scatterplot3d(clean_data$PRE_3P_PCT, clean_data$PRE_FT_PCT, clean_data$NBA_FG3_PCT,
                     color = colors, pch = 16,
                     xlab = "Pre-3P%", ylab = "Pre-FT%", zlab = "NBA 3P%",
                     main = "Colored by PRE_COL_OR_INT")
s3d$plane3d(model, lty.box = "solid")

legend("topright", legend = levels(color_factor),
       col = rainbow(length(levels(color_factor))), pch = 16)

```



MLR of both PRE-NBA FT% and 3P% vs NBA 3P% filtered for 25 attempts

```

data_filtered2 <- data %>%
  filter(PRE_3PA >= 25 & PRE_FTA >= 25 & NBA_FG3A >= 25)

data_filtered2 <- normalize_stats(data_filtered2)

clean_data <- na.omit(data_filtered2[c("PLAYER", "NBA_FG3_PCT", "PRE_3P_PCT", "PRE_FT_PCT", "PRE_COL_or_INT")])
model <- lm(NBA_FG3_PCT ~ PRE_3P_PCT + PRE_FT_PCT, data = clean_data)
summary(model)

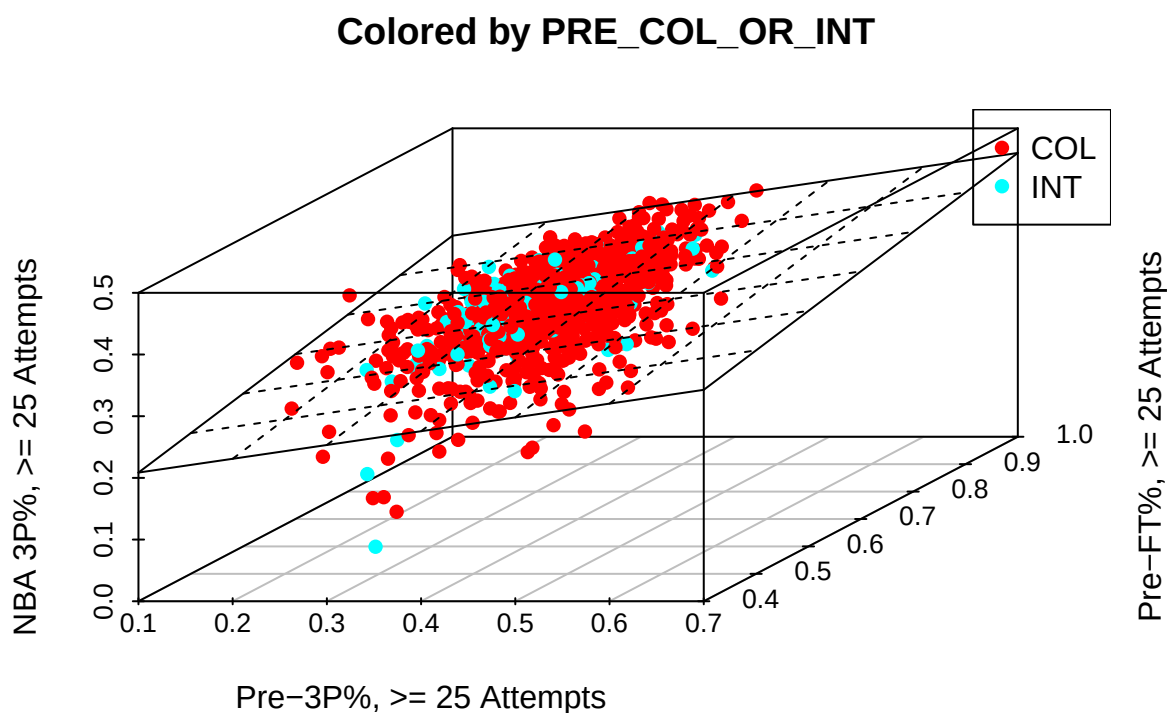
```

##


```
## Call:
## lm(formula = NBA_FG3_PCT ~ PRE_3P_PCT + PRE_FT_PCT, data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.252344 -0.019491  0.008559  0.033728  0.095787
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.10796    0.01583   6.822 1.49e-11 ***
## PRE_3P_PCT   0.22357    0.03315   6.743 2.51e-11 ***
## PRE_FT_PCT   0.19565    0.02277   8.593 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04865 on 1085 degrees of freedom
## Multiple R-squared:  0.1633, Adjusted R-squared:  0.1618
## F-statistic: 105.9 on 2 and 1085 DF,  p-value: < 2.2e-16
```

```
color_factor <- as.factor(clean_data$PRE_COL_or_INT)
colors <- rainbow(length(levels(color_factor)))[color_factor]
s3d <- scatterplot3d(clean_data$PRE_3P_PCT, clean_data$PRE_FT_PCT, clean_data$NBA_FG3_PCT,
                     color = colors, pch = 16,
                     xlab = "Pre-3P%, >= 25 Attempts", ylab = "Pre-FT%, >= 25 Attempts", zlab = "NBA 3P%",
                     main = "Colored by PRE_COL_OR_INT")
s3d$plane3d(model, lty.box = "solid")

legend("topright", legend = levels(color_factor),
       col = rainbow(length(levels(color_factor))), pch = 16)
```



```
FTto3 <- lm(NBA_FG3M ~ PRE_FT_PCT, data = data_filtered)
summary(FTto3)
```

```
##
## Call:
## lm(formula = NBA_FG3M ~ PRE_FT_PCT, data = data_filtered)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.052067 -0.015470 -0.001225  0.014015  0.073206
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.064028   0.005776  -11.09  <2e-16 ***
## PRE_FT_PCT   0.142273   0.007799   18.24  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02109 on 1184 degrees of freedom
## Multiple R-squared:  0.2194, Adjusted R-squared:  0.2187
## F-statistic: 332.8 on 1 and 1184 DF,  p-value: < 2.2e-16
```

```
ggplot(data_filtered, aes(x = PRE_FT_PCT, y = NBA_FG3M)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", se = TRUE, color = "red", alpha = 0.2) +
```

```
labs(
  x = "Pre-NBA Free Throw Percentage",
  y = "NBA 3-Point Made Per Game",
  title = "Free Throw % vs 3-Point Made Per Minute (Min. 25 Attempts Each)",
  subtitle = paste("n =", nrow(data_filtered), "players")
) +
theme_minimal() +
theme(plot.title = element_text(hjust = 0.5),
      plot.subtitle = element_text(hjust = 0.5))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

